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Assessing the cumulative impact of alcohol marketing on young people's drinking: Cross-sectional data findings

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As alcohol marketing remains a highly debated and politically charged issue, we examine the cumulative impact of alcohol marketing on alcohol initiation and drinking behaviour among youth (12–14 years). Cross-sectional data come from a cohort of 920 second year school pupils from Scotland. Regression models, with multiple control variables, were employed to examine the relationship between awareness of, and involvement with, a range of alcohol marketing communications, and drinking behaviour and intentions. Marketing variables were constructed for 15 different types of alcohol marketing, including marketing in new media. Drinking behaviour measures included drinking status and future drinking intentions. Significant associations were found between awareness of, and involvement with, alcohol marketing and drinking behaviour and intentions to drink alcohol in the next year. Given these associations, our study suggests the need for a revision of alcohol policy: one limiting youth exposure to these seemingly ubiquitous marketing communications.

Keywords Cross-sectional, survey, alcohol, marketing, adolescents

INTRODUCTION

Alcohol consumption in the UK has been steadily increasing since the mid-1990s (HMRC, 2007), with this trend not confined to adults but extending to youth as well. Indeed, the amount of alcohol that young

people consumed increased between 2000 and 2006, with increases of 43.4% for 11–13-year-old-boys and 82.6% for 11–13-year-old girls (Office for National Statistics, 2007). And it is not just the amount of alcohol that youths drink that is a cause for concern, but also the frequency, and concomitant harm. For instance, for 15–16 year olds in the UK, the estimated average level of alcohol last consumed, levels of binge drinking and associated harm are among the highest in Europe (Hibell et al., 2009).

These alarming trends are invariably accompanied by multiple health and social problems, which elicit high levels of public concern (HM Government, 2007). For example, a 20% increase in hospital admissions among youth due to alcohol use has been reported (HES, 2007) and nearly half of all 10–17-year-old regular drinkers have admitted to some sort of criminal activity or disorderly behaviour (Home Office, 2004). Also, in the longer term, behaviours such as binge drinking among young people is a predictor of alcohol dependency in later life (Jefferis, Power, & Manor, 2005) and is associated with coronary heart disease, liver cirrhosis and stroke (Britton & McPherson, 2001; Gutjahr, Gmel, & Rehm, 2001; Leon & McCambridge, 2006).

The extent and nature of alcohol harm in the UK has stimulated a proliferation of research into risk factors potentially influencing adolescent drinking behaviour. One factor identified is that of alcohol marketing (Booth et al., 2008). The alcohol market in the UK is considerable, estimated to be worth in excess of 41.6 billion pounds in 2007 (Keynote, 2008), with the trend towards consolidation in the alcohol market leading to the emergence of global brands with colossal marketing budgets. This is exemplified by the estimated 300

million pounds spent annually on alcohol advertising in the UK (WARC, 2006); with total marketing spend surpassing 800 million pounds per annum (Prime Minister's Strategy Unit, 2003). The latter figure shows that much alcohol marketing goes beyond conventional marketing channels, such as television (TV), print media and outdoor advertising, to 'below the line' channels such as sponsorship, new media and viral marketing (Jernigan, 2001). However, there is currently a gap in research focusing on these and new electronic forms of alcohol marketing.

Research assessing alcohol marketing and drinking behaviour has taken two principal forms: econometric and consumer research. Econometric studies analyse the relationship between overall levels of alcohol consumption, typically using sales data, and overall levels of advertising or marketing expenditure. Most econometric studies suggest alcohol marketing to have little or no effect on overall consumption levels (Duffy, 1991; Nelson, 2003). However, econometric studies typically suffer from a number of methodological weaknesses (Hastings, Anderson, Cooke, & Gordon, 2005) and, importantly, provide little understanding of segmented market groups, such as young people. Consumer studies use individuals as the unit of analysis and examine how knowledge, attitudes and behaviour are influenced by exposure to alcohol marketing. Unlike most econometric data, consumer research does suggest that alcohol marketing influences drinking behaviour, with recent systematic reviews of the alcohol marketing literature confirming this and moving the debate on from one of plausibility to veracity (Booth et al., 2008; Anderson, de Bruijn, Angus, Gordon & Hastings, 2009; Smith & Foxcroft, 2009). Research in the US also suggests that youth are overexposed to alcohol marketing, indicating that the volume, as well as the nature and content of marketing necessitates some attention (Jernigan, Ostroff, Ross, Naimi, & Brewer, 2007; CAMY, 2008).

Our research addresses two important gaps in the literature. First, it assesses the cumulative impact of all alcohol marketing communications on adolescents' current drinking and future drinking intentions by examining new as well as conventional media. Second, since most consumer research examining the relationship between adolescent drinking and alcohol marketing has been conducted in the US, this study presents much needed research assessing this relationship in the UK. Our research hypotheses are such that the more aware and appreciative of alcohol marketing that young people are, the more likely they are to have consumed alcohol (H1), and the more likely they are to think that they will drink alcohol in the next year (H2).

METHODS

Design

Cross-sectional data are used from the *Assessing the Cumulative Impact of Alcohol Marketing on Youth*

Drinking study, a longitudinal questionnaire-based design adapted from a tobacco marketing study in the UK (MacFadyen, Hastings, & MacKintosh, 2001). The questionnaire was informed by a preliminary research phase, which included an audit of alcohol marketing communications, in-depth interviews with stakeholders and qualitative focus groups with 13–15 year olds (Gordon, Hastings, Moodie, & Eadie, 2010). Questionnaire design and development reflected findings from this earlier research phase and involved an extensive, iterative piloting process to ensure the acceptability, appropriateness and comprehension of questionnaire content, and also that respondents could accurately reflect their views and experiences without constraining responses that can be caused due to not understanding questions or response categories (Kinnear & Taylor, 1991; Fink & Kosecoff, 1998). Two piloting procedures were used, focus groups with four or five respondents per group ($n=16$) and observed interviews ($n=12$). These were conducted in two stages to guide questionnaire content and comprehension. The final survey involved an interview-administered questionnaire measuring awareness, appreciation and involvement with alcohol marketing, and a self-completion questionnaire measuring drinking and associated behaviours.

SAMPLE

An information pack was sent out to the homes of all second year (12–14 years, mean age 13 years) pupils attending schools in three local authority areas in the West of Scotland. These packs contained an information sheet and parental and respondent consent forms, with an incentive of a small gift token offered for participation. This generated a sample of 920 respondents. In terms of sample characteristics, the achieved sample was relatively evenly distributed by social grade based on the occupation of the head of the household [46.3% ($n=426$) ABC1 (middle class) and 53.3% ($n=491$) C2DE (working class)] (Wilmshurst & MacKay, 1999) and gender [52.9% ($n=487$) female and 47.1% ($n=433$) male]. Sample ethnicity was predominantly white 93.3% ($n=858$), with 3.3% ($n=30$) identifying themselves as Asian, 1.3% mixed race ($n=12$), 1.2% black ($n=11$), 0.1% Chinese ($n=1$) and 0.4% other ($n=4$). For religious identification, most of the sample were Christian 62.4% ($n=574$) or had no religiosity 32% ($n=294$), with 3.7% Muslim ($n=34$), 0.5% Sikh ($n=5$), 0.2% Hindu ($n=2$), 0.1% Buddhist ($n=1$) and 0.4% other ($n=4$).

PROCEDURE

The fieldwork comprised face-to-face interviews conducted in-home, by professional market researchers, accompanied by a self-completion questionnaire to gather more sensitive data on alcohol behaviour. Confidentiality and anonymity of personal data were

assured. Parental or guardian permission and participant consent were secured prior to each interview. To limit socially desirable responses, due to parental or guardian presence and given the sensitive nature of the research topic, numbered showcards were used for much of the interviewer-administered questionnaire. Alcohol consumption was measured via a self-completion questionnaire, which the respondents were instructed to place in a sealed envelope following completion and hand back to the researcher at the end of the interview. Ethical approval was obtained from the University of Stirling's Marketing Department Ethics Committee and interviewers adhered to the Market Research Society code of conduct.

MEASURES

Demographic information

Demographic control variables measured included *age*, *gender*, *social grade* (based upon occupation of head of household); *ethnicity* and *religion*.

Drinking behaviour and future drinking intentions

Drinking status was assessed with the question: 'Have you ever had a proper alcoholic drink—a whole drink, not just a sip?' Those who had tried a whole alcohol drink, and not just a sip, were classified as drinkers, and those who had not done so as non-drinkers. *Future drinking intentions* were assessed via a single item asking: 'Do you think you will drink alcohol at any time during the next year', with the following response categories; 'definitely not', 'probably not', 'probably yes', 'definitely yes' and 'I'm not sure'.

Initiation into drinking was recorded by asking respondents: *age of first alcoholic drink* (in years). The *number of alcoholic units last consumed* was calculated by estimating the amount in millilitres of each type of alcoholic drink consumed and the ABV of each drink, based on responses to the following: type(s) of alcohol consumed (e.g. beer, wine, vodka), drinking vessel used (recorded using a visual), the amount of each drink consumed (more than one full bottle/can/glass, one full bottle/can/glass, $\frac{3}{4}$, $\frac{1}{2}$, $\frac{1}{4}$ or less than $\frac{1}{4}$ of a bottle/can/glass).

Awareness of alcohol marketing

Alcohol marketing awareness was assessed across 15 types of marketing, identified from formative research (Gordon et al., 2010), which were grouped using principal components analysis. Respondents were asked if they had seen any alcohol marketing in each of the channels listed (Yes/No/Do not Know) and if so what brands they could recall seeing marketed in this way. Respondents were also given the opportunity to record any other marketing channels not mentioned.

Involvement in alcohol marketing

To assess *involvement* in alcohol marketing, respondents were asked whether they had: received free

samples of alcohol products, received free gifts showing alcohol brand logos, received special price offers for alcohol, received promotional mail or e-mails mentioning alcohol brands, owned clothing or other alcohol-branded items, looked at websites for alcohol brands, downloaded mobile phone or computer screen-savers featuring alcohol brands and used social networking sites containing alcohol brands or logos (Yes/No/Do not know).

Control variables

A range of control variables, suggested in the literature, were included in the survey and the analyses. Parental attitudes towards drinking and alcohol consumption were assessed using the following measures: drinking behaviour (whether mother or father drank, or they did not have/see their mother or father); perceived parental drinking approval (whether they thought that their parents would think it was OK to drink alcohol to see what it was like, or they didn't know). Similar questions were asked with respect to siblings' drinking behaviour and perceived approval, and also peers' drinking behaviour and perceived approval. Respondents were also asked, as far as they knew, how many of their friends drank alcohol at least once a week ('all of them', 'most of them', 'about half of them', 'a few of them', 'none of them' or 'I'm not sure'). The latter variables were re-coded as categorical to maximise the sample size for the statistical analyses by retaining cases with responses of 'do not know' or 'not stated' (the categories were 'few or none', 'about half/most/all' and 'do not know/not stated'). Liking of school ('dislike school a lot', 'dislike school a little', 'neither like nor dislike school', 'like school a little' or 'like school a lot') and rating of school work compared to other pupils ('a lot worse', 'a bit worse', 'about the same', 'a bit better' or 'a lot better') were also assessed. Liking of adverts in general and alcohol adverts in particular were also assessed ('I like (alcohol) adverts a lot', 'I like (alcohol) adverts a little', 'I neither like nor dislike (alcohol) adverts', 'I dislike (alcohol) adverts a little' and 'I dislike (alcohol) adverts a lot'). Finally, age was included as a control variable because the year group included adolescents ranged between 12 and 14 years and bivariate analysis revealed a significant association between age and drinking status.

ANALYSES

Bivariate analyses were conducted to examine relationships between the data and inform the multivariate analyses. Principal components analyses were performed on (1) whether sibling(s)/parents/friends consider it OK to try drinking to see what it is like, (2) types of alcohol communications items and (3) involvement in alcohol marketing.¹ The principal components were extracted using varimax rotation with the criteria of having eigenvalues greater than 1

and factor loadings greater than 0.4, together with consideration of the scree plot. Composite scores were computed, based on the factor loadings, and used in the subsequent logistic and linear regression analyses.

Two logistic regressions were performed with drinking status as the dependent variable. The two models examined the association between (1) drinking status and the number of marketing channels aware of and (2) drinking status and types of alcohol marketing aware of and involvement in alcohol marketing. In each, a wide range of other variables were controlled for. The choice of control variables was informed by both the literature and earlier bivariate analysis. Variables were entered in the following blocks using forward likelihood ratio: (1) how many of their friends drink, whether their mother, father and sibling(s) drink and perceived prevalence of drinking among 13–15 year olds; (2) demographics (age, sex, social grade, ethnicity and religion); (3) whether they thought sibling(s)/parents/friends would consider it OK to try drinking to see what it is like; (4) liking of school, rating of school work, liking of adverts and liking of alcohol adverts; and (5) the number of alcohol marketing channels respondents were aware of (in the first analysis), types of alcohol marketing communications respondents were aware of and involvement in alcohol marketing (in the second analysis).

Stepwise linear regression analyses were then performed with drinking intentions within the next year as the dependent variable and the following independent variables: the control variables used in the logistic regressions, the number of alcohol marketing channels respondents were aware of (linear regression analysis 1), types of alcohol communications respondents were aware of (linear regression analysis 2) and involvement in alcohol marketing. The levels of significance for including and removing a variable were 0.05 and 0.10, respectively.

RESULTS

Drinking behaviour

Of the 318 drinkers, i.e. those who reported having had a proper alcoholic drink, the mean age at which they reported having their first alcoholic drink was 11.1 years ($SD = 1.7$) and the mean number of alcoholic units consumed the last time they drank was 4.6 ($SD = 5.3$).

ALCOHOL MARKETING AWARENESS

Table I illustrates the respondents' awareness of alcohol marketing channels. For the entire sample, awareness was highest for TV advertising (77%), branded clothing (66%), sport sponsorship (61%) and price promotions (60%). For electronic communications, approximately a quarter of the sample (24%) were aware of mobile phone/computer screensavers.

Drinkers indicated higher awareness of alcohol marketing in many of the channels including TV/cinema, newspapers/magazines, in-store, price promotions, sports sponsorship, screensavers, social networking sites and unusual product design (Table I).

INVOLVEMENT IN ALCOHOL MARKETING

The most common type of alcohol marketing involvement was ownership of alcohol-branded clothing (45%), free branded gifts (11%) and price promotions (10%) (Table II). Significantly more drinkers reported involvement in each form of alcohol marketing and, correspondingly, more non-drinkers reported no involvement in any form of alcohol marketing.

ASSOCIATIONS WITH DRINKING STATUS

The first logistic regression model examines the association between drinking status and the number of marketing channels the adolescents were aware of, after controlling for the effects of the control variables listed in the section 'Measures'.

It was found that being aware of more alcohol marketing channels and liking alcohol advertisements increased the odds of being a drinker by 12% and 31%, respectively, see Table III. Having sibling(s) who drink alcohol, believing that at least half their friends drink alcohol and perceiving that parents/sibling(s)/friends would consider it OK to try drinking alcohol to see what it is like, also increased the odds of being a drinker.² The odds of being a drinker were decreased with greater liking of school.³

In the second logistic regression model the total number of alcohol marketing channels of which adolescents were aware was replaced by the types of alcohol marketing channels of which they were aware and involvement in alcohol marketing was also included.

Greater awareness of advertisements and promotions and greater liking of alcohol adverts increased the odds of being a drinker by 137% and 28%, respectively, see Table IV. However, the confidence interval for awareness of advertisements and promotions was very wide. Involvement in electronic marketing (including having looked at a website for alcohol brands or about drinking; downloaded a mobile phone or computer screensaver containing an alcohol brand name or logo; and used a web home page) increased the odds of being a drinker by 300%. However, the confidence interval for electronic marketing was also extremely wide. Having siblings who drink, having friends (half or more) who drink alcohol, and perceiving that their parents/siblings/friends would consider it acceptable to try alcohol to see what it is like also increased the odds of being a drinker. Having a better liking of school decreased the odds of being a drinker.

Table I. Adolescents' awareness of alcohol marketing, by drinking status.

		Total N = 920		Drinkers N = 318		Non-drinkers N = 594		P-value (χ^2 test)
Alcohol channels*	Marketing	% (valid)	N	% (valid)	N	% (valid)	N	
	<i>Ads and promotions</i>							
1	TV/cinema	77	707	82	262	74	439	<0.01
2	Posters/billboards	53	488	56	178	52	307	n.s.
3	Newspapers/magazines	32	298	39	123	29	173	<0.01
4	In-store	55	510	62	196	52	311	<0.01
5	Price promotions	60	554	67	214	57	336	<0.01
	<i>Sports-related</i>							
6	Sports sponsorship	61	565	68	217	58	343	<0.01
7	Clothing	66	607	70	223	64	381	n.s.
	<i>Electronic communications</i>							
8	E-mail	5	47	7	21	4	26	n.s.
9	Websites	7	60	7	22	6	37	n.s.
10	Mobile Phone/computerscreensaver	24	224	31	99	21	124	<0.01
11	Social networking sites	12	113	22	70	7	43	<0.001
	<i>Arts-related</i>							
12	Music sponsorship	34	314	34	107	35	205	n.s.
13	TV/film sponsorship	30	278	33	105	29	170	n.s.
14	Celebrity endorsement	15	136	17	54	14	82	n.s.
15	Product design	20	180	24	76	17	103	<0.05
Mean number (SD) of marketing channels aware of**		5.5 (2.8)		6.2 (2.77)		5.2 (2.7)		

Notes: n.s., not significant.

*Categorisation of alcohol marketing communication channels was derived from a principal components analysis reported under multivariate analyses.

**One way analysis of variance, $F_{(1, 910)} = 27.511$, $p < 0.001$.

Table II. Type of promotional activity participated in, or used by, drinkers and non-drinkers.

		Total N = 920		Drinkers N = 318		Non-drinkers N = 594		P-value (χ^2 test)
Promotions		% (valid)	N	% (valid)	N	% (valid)	N	
Free samples		3	29	6	19	2	10	<0.001
Free branded gifts		11	101	14	45	9	56	<0.05
Price promotions		10	90	13	42	8	47	<0.05
Promotional mail/e-mails		8	71	11	34	6	36	<0.05
Branded clothing		45	418	51	162	43	254	<0.05
Websites		4	41	7	23	3	18	<0.01
Mobile phone/computer screensaver		7	64	10	33	5	31	<0.01
Social networking sites		8	72	14	45	5	27	<0.001
None of these		40	364	31	100	44	261	<0.001

FUTURE DRINKING INTENTIONS

Of the 912 respondents who reported their intentions concerning drinking alcohol in the next year, 29% ($n = 262$) indicated that they would definitely not drink, 25% ($n = 231$) that they would probably not, 16% ($n = 150$) that they were not sure, 23% ($n = 212$) that they probably would and 6% ($n = 57$) that they definitely would drink alcohol in the next year.

The first linear regression model, examining the association between future drinking intentions and awareness of number of alcohol marketing channels, included 860 respondents, was significant ($F = 42.660$, $df = 13, 847$, $p < 0.001$) and accounted for 39% of the variance. The more alcohol marketing channels the respondents were aware of and the better their liking of alcohol advertisements, the more likely they were to think that they would drink alcohol in the next year.

Table III. Logistic regression of association between awareness (amount of) and appreciation of alcohol marketing and drinking status.

		<i>N</i>	Odds ratio	95.0% CI for odds ratio		Significance
Sibling drinking						<0.05
	Sibling(s) do not drink	419	1.00			
	Sibling(s) drink	268	1.667	1.140	2.438	<0.01
	No sibling(s)	128	1.120	0.686	1.827	n.s.
	Do not know if sibling(s) drink	49	0.872	0.418	1.819	n.s.
Close friends drinking						<0.001
	Few or none drink	584	1.000			
	About half/most/all drink	127	3.710	2.322	5.928	<0.001
	Not sure/not stated	153	0.971	0.622	1.515	n.s.
Mother's drinking						n.s.
	Mother does not drink	287	1.000			
	Mother drinks	496	1.277	0.870	1.874	n.s.
	Not sure/not stated	73	1.205	0.628	2.312	n.s.
	No mother/do not see mother	8	1.265	0.188	8.512	n.s.
Ethnic group						
	White	808	1.00			
	Asian or Asian British/mixed/other	56	0.516	0.198	1.346	n.s.
Perceptions of others' views on trying alcohol (–ve = not OK, +ve = OK)		864	1.431	1.316	1.556	<0.001
Liking of school (1 = dislike a lot, 5 = like a lot)		864	0.832	0.726	0.954	<0.01
Liking of alcohol adverts (1 = dislike a lot, 5 = like a lot)		864	1.307	1.110	1.538	<0.01
Number of forms of alcohol marketing aware of		864	1.116	1.049	1.188	<0.01

Notes: Hierarchical logistic regression controlled for Block 1, drinking among siblings, close friends, mother and father, perceptions of drinking prevalence among 13–15 year olds; Block 2, gender, age, social grade, ethnic group, religion; Block 3, perceptions of siblings/parents/friends' views on whether OK to try drinking; Block 4, liking of school, perception of own school work relative to others in their year, liking of adverts in general, liking of alcohol adverts; and Block 5, amount of alcohol marketing aware of.

864 cases analysed, 56 missing cases.

Nagelkerke $R^2 = 0.341$.

Cases correctly classified = 76.5%; 54% of drinkers and 88% of non-drinkers were correctly classified.

Dependent variable = Drinking status (1 = had a proper drink, 0 = not had a proper drink).

Base: All respondents.

n.s., not significant.

(Table V). A greater intention to drink was also significantly associated with greater perception that others consider it OK to try drinking, having at least half of their friends drinking, having siblings who drink (compared with siblings who do not drink) and being non-religious (compared with being Christian). A greater liking of school was associated with lower intention to drink alcohol in the next year (Table V).

The second linear regression model, examining the association between future drinking intentions and types of alcohol marketing channel, included 860 respondents, was significant ($F = 39.795$, $df = 14$, 846, $p < 0.001$) and accounted for 39% of the variance. Better liking of alcohol advertisements and greater awareness of alcohol advertising and promotions (including adverts on TV, in the cinema, newspapers or magazines, posters/billboards, signs in shops and special price offers) were each significantly associated with stronger intention to drink alcohol in the next year.

Additionally, involvement in electronic marketing was significantly associated with greater intention to

drink alcohol in the next year (Table VI). As in the first linear regression, greater intention to drink was also significantly associated with greater perception that others consider it OK to try drinking, having at least half of their friends drinking, having siblings who drink (compared with siblings who do not drink), and being non-religious (compared with being Christian). A greater liking of school was associated with lower intention to drink alcohol in the next year.

DISCUSSION

The findings support both hypotheses as awareness and appreciation of alcohol marketing was significantly associated with both being a drinker and intending to drink within the next year; consistent with the extant literature. We include channels for which there is very little research such as new media and different types of alcohol sponsorship, with most research typically focusing on advertising in conventional media (TV, radio, print) (Snyder, Fleming-Milici, Slater,

Table IV. Logistic regression of association between awareness (types of) and appreciation of alcohol marketing and drinking status.

		N	Odds ratio	95.0% CI for odds ratio		Significance
Sibling drinking						<0.05
	Sibling(s) do not drink	419	1.00			
	Sibling(s) drink	268	1.651	1.126	2.421	<0.05
	No sibling(s)	128	1.134	0.693	1.855	n.s.
	Do not know if sibling(s) drink	49	0.798	0.373	1.707	n.s.
Close friends drinking						<0.001
	Few or none drink	584	1.000			
	About half/most/all drink	127	3.612	2.252	5.793	<0.001
	Not sure/not stated	153	0.989	0.633	1.543	n.s.
Mother's drinking						n.s.
	Mother does not drink	287	1.000			
	Mother drinks	496	1.323	0.897	1.952	n.s.
	Not sure/not stated	73	1.284	0.669	2.466	n.s.
	No mother/do not see mother	8	1.355	0.201	9.128	n.s.
Ethnic group						
	White	808	1.00			
	Asian or Asian British/mixed/other	56	0.488	0.182	1.309	n.s.
Perceptions of others' views on trying alcohol (–ve = not OK, +ve = OK)		864	1.439	1.323	1.565	<0.001
Liking of school (1 = dislike a lot, 5 = like a lot)		864	0.834	0.727	0.958	<0.05
Liking of alcohol adverts (1 = dislike a lot, 5 = like a lot)		864	1.279	1.084	1.508	<0.01
Awareness of alcohol advertising (in TV/cinema, press, outdoor such as billboards, in-store and special price offers)		864	2.374	1.301	4.333	<0.01
Participation in electronic alcohol marketing		864	4.000	1.485	10.776	<0.01

Notes: Hierarchical logistic regression controlled for Block 1, drinking among siblings, close friends, mother and father, perceptions of drinking prevalence among 13–15 year olds; Block 2, gender, age, social grade, ethnic group, religion; Block 3, perceptions of siblings/parents/friends' views on whether OK to try drinking; Block 4, liking of school, perception of own school work relative to others in their year, liking of adverts in general, liking of alcohol adverts; and Block 5, forms of alcohol marketing aware of and participated in.

864 cases analysed, 56 missing cases.

Nagelkerke $R^2 = 0.351$.

Cases correctly classified = 77%; 54.5% of drinkers and 89% of non-drinkers were correctly classified.

Dependent variable = Drinking status (1 = had a proper drink, 0 = not had a proper drink).

Base: All respondents.

n.s., not significant.

Sun, & Strizhakova, 2006) or alcohol-branded merchandise (Fisher, Williams Miles, Austin, Camargo, & Colditz, 2007; McClure, Stoolmiller, Tanski, Worth, & Sargent, 2009) in isolation. Indeed, of the 13 studies included in a recent systematic review (Anderson et al., 2009), only 2 examined the cumulative impact of alcohol marketing across several channels. More specifically, Ellickson, Collins, Hambarsoomians, and McCaffrey (2005) considered TV and magazine advertising and in-store promotions, and Collins, Ellickson, McCaffrey, and Hambarsoomians (2007) considered TV, radio and magazine advertising, in-store promotions and ownership of branded merchandise. Our research, albeit cross-sectional in nature, therefore helps to illustrate the extent, nature and reach of contemporary alcohol marketing.

The findings also help reveal the breadth of alcohol marketing channels that young people are aware of, or involved with. Importantly, for many of the forms of

alcohol marketing, drinkers had higher awareness than non-drinkers. That almost half the sample had alcohol-branded clothing (45%) is likely due to ownership of alcohol branded football shirts, with the two major football teams in Scotland sponsored by Carling; who paid many millions of pounds for the privilege. Exemplifying the value of sports sponsorship, which according to internal tobacco industry documents was more highly prized than some forms of mass media advertising (Anderson, Dewhirst, & Ling, 2006). Carling generated the highest level of awareness among our respondents and indeed has now become the most popular lager brand in the UK (Euromonitor, 2008). Coors, who own Carling, has taken the decision to voluntarily remove the name from children's football shirts 2 years ahead of the deadline set by the Portman Group to do so. However, two important points must be noted here. First, having established the most popular UK alcohol brand Coors have already

Table V. Association between the number of alcohol marketing channels adolescents were aware of and their reported intention to drink alcohol in the next year.

	Unstandardised coefficients		Standardised coefficients		<i>t</i>	Significance
	<i>B</i>	Standard error	Beta			
Constant	−0.178	1.087			−0.164	n.s.
Perceptions of others' views on trying alcohol (−ve = not OK, +ve = OK)	0.240	0.018	0.393		13.568	<0.001
About half/most or all friends drink	0.467	0.102	0.129		4.594	<0.001
Sibling(s) drink(s)	0.345	0.081	0.124		4.250	<0.001
Does not have siblings	0.081	0.103	0.022		0.781	n.s.
Mother drinks	0.153	0.087	0.059		1.768	n.s.
Does not know if mother drinks	0.206	0.134	0.044		1.529	n.s.
Father drinks	0.130	0.083	0.049		1.556	n.s.
Age	0.156	0.083	0.051		1.885	n.s.
Ethnic group: non-white	−0.259	0.149	−0.049		−1.742	n.s.
Religion: none	0.226	0.075	0.082		3.013	<0.01
Liking of school (1 = dislike a lot, 5 = like a lot)	−0.094	0.029	−0.090		−3.219	<0.01
Liking of alcohol ads(1 = dislike a lot, 5=like a lot)	0.156	0.034	0.127		4.564	<0.001
Number of channels of alcohol marketing aware of	0.054	0.013	0.118		4.290	<0.001

Notes: Dependent variable = Intention to drink alcohol in the next year (1 = definitely will not, 5=definitely will drink).

Base: All respondents.

n.s., not significant.

Table VI. Association between the types of alcohol marketing channels adolescents were aware of and their reported intention to drink alcohol in the next year.

	Unstandardised coefficients		Standardised coefficients		<i>t</i>	Significance
	<i>B</i>	Standard error	Beta			
Constant	0.012	1.084			0.011	n.s.
Perceptions of others' views on trying alcohol (−ve = not OK, +ve = OK)	0.241	0.018	0.395		13.662	<0.001
About half/most or all friends drink	0.456	0.102	0.126		4.475	<0.001
Sibling(s) drink(s)	0.341	0.081	0.122		4.204	<0.001
Does not have siblings	0.086	0.103	0.024		0.837	n.s.
Mother drinks	0.158	0.087	0.061		1.825	n.s.
Does not know if mother drinks	0.216	0.135	0.047		1.605	n.s.
Father drinks	0.133	0.083	0.051		1.603	n.s.
Age	0.147	0.083	0.048		1.778	n.s.
Ethnic group: non-white	−0.274	0.149	−0.052		−1.845	ns
Religion: none	0.224	0.075	0.081		2.994	<0.01
Liking of school (1 = dislike a lot, 5 = like a lot)	−0.094	0.029	−0.090		−3.225	<0.01
Liking of alcohol ads(1 = dislike a lot, 5=like a lot)	0.149	0.034	0.121		4.328	<0.001
Awareness of alcohol advertising (in TV/cinema, press, outdoor such as billboards, in-store and special price offers)	0.373	0.121	0.085		3.079	<0.01
Participation in electronic alcohol marketing	0.552	0.211	0.074		2.610	<0.01

Notes: Dependent variable = Intention to drink alcohol in the next year (1 = definitely will not, 5 = definitely will drink).

Base: All respondents.

n.s., not significant.

achieved their primary goal and therefore the removal of the Carling brand name from the Rangers and Celtic team shirts is of little value to those youth who associate Carling, and perhaps alcohol in general, with sport, glamour and wealth. Second, Coors only elected

to remove this sponsorship following the Portman Group's decision to do so, which suggests the use of stakeholder marketing, i.e. portraying the company as a responsible organisation concerned with public health, has replaced the more overt marketing

communications that enabled the brand to be so successful (Baggot, 2006).

Among the young people in the study, 1 in 10 indicated that they had participated in price promotions for alcohol, which raises further concerns given that our sample was between 4 and 6 years below the legal age of 18 for the purchase of alcohol. More robust off-sales licensing regulations and test purchasing schemes are to be introduced by the Licensing (Scotland) Act 2005 in September 2009, and could serve as useful countermands. The Scottish Government has also published plans to introduce minimum alcohol unit pricing which may limit alcohol promotions (Scottish Government, 2009).

That participation in electronic alcohol marketing communications was associated with drinking, albeit with a small effect size, is illuminating given the dearth of research examining alcohol marketing in new media channels, which are less well regulated than more conventional marketing channels (British Medical Association – Board of Science, 2009). These findings provide some disquiet when it is considered that 96% of almost 9000 youth in England reported having used the Internet in the past week, according to the recent research assessing youth gambling behaviour (MORI, 2009).

Our findings also demonstrated the importance of peer and parental influence as control variables. Peer and parental influence were found to be significant in each of our regression analyses, and this is consistent with the literature examining alcohol marketing and drinking behaviour (Petraitis, Flay & Miller, 1995; Szalay, Inn, & Doherty, 1996).

There are some study limitations that must be considered. First, cross-sectional findings cannot tell us about any impact from marketing on changes in drinking behaviour over time, or indeed drinking initiation. Although our results suggest an association between awareness of, and exposure to, marketing and drinking behaviour, it cannot assess causality. Second, the relatively small sample size and locality of the study may affect the generalisability of the findings. Third, despite including a large number of potential confounding effects in our survey, there may be other variables of which we are not aware, or have not been measured. However, the findings are consistent with the existing evidence (Booth et al., 2008) and we will be able to assess causality at follow-up.

CONCLUSION

With the current regulatory system focused towards controlling content, our findings suggest the more pertinent issue to be level of exposure. Given that the evidence base suggests an association, and indeed a causal link, between alcohol marketing and youth drinking, the current regulation does not seem to afford adolescents adequate protection from alcohol marketing exposure (Anderson, 2009). This suggests that the

regulation of alcohol marketing, especially new media channels requires serious examination.

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NOTES

1. It is considered legitimate to perform principal components analysis on dichotomous variables if the underlying correlation matrix consists of biserial correlations.
2. Due to the large number of variables included in the statistical analyses, missing data and some 'do not know' responses reduced the sample sizes. To maximise the numbers of respondents represented in the analyses, 'do not know' responses were included as categories in the analyses where possible.
3. The interpretation of some of the results might appear counter-intuitive in relation to whether the OR values are less than or greater than 1, due to the direction in which individual variables were coded in the questionnaire.
4. The National Preventive Research Initiative (NPRI) is an initiative to support high quality research aimed at identifying effective approaches to reduce risk factors and influence health behaviour in order to positively impact upon the incidence of new cases of major preventable diseases. The initiative is supported by a consortium of major research bodies and charities including the Economic and Social Research Council; Medical Research Council; British Heart Foundation; Cancer Research UK; Department of Health; Diabetes UK; Food Standards Agency; Research and Development Office for the Northern Ireland Health and Social Services; Chief Scientist Office, Scottish Executive Health Department; Wales Office of Research & Development; and World Cancer Research Fund.

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